

Brandon Ho

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EDUCATION

University of California San Diego

GPA: 3.93/4 (M.S.)

B.S. Computer Engineering (Magna Cum Laude), M.S. Intelligent Systems, Robotics & Control

June 2024

- **Relevant Coursework:** Software Engineering, Deep Learning, Operating Systems, Computer Vision, Autonomous Vehicles, Sensing & Estimation, Planning & Learning, Robot Sys Design & Implementation, Statistical Learning

EXPERIENCE

Relativity Space

Long Beach, CA

Robotics Software Engineer II

September 2024 - Current

- Building a flexible, **real-time** robotics software platform (C++) to support large metal additive manufacturing
- **Project Management:** Successfully executed **1-year R&D project** to revamp core SW middleware with team of 3 engineers, wrote and iterated on roadmap, presented monthly to stakeholders, delegated work via **Jira**
- **RT Middleware:** Built novel RT-capable **Zenoh+EVL**-based RMW for in-band/out-of-band comms; migrated platform from shm/DDS, increasing throughput **10x**, reliability **1.5x**, reducing codebase redundancy **15%**
- **RT Linux Kernel:** Automated **Xenomai4** build/packaging via **Ansible**; contributed upstream patches fixing network stack packet leak and page faults in real-time processes, **reducing core SW crashes by 90%**
- **Sensor Integration:** Integrated Gocator 2D laser profilometer with real-time **RANSAC circle fitting** to extract weld bead geometry at **10Hz**, increasing part geometry **data completeness 20x**, reducing noise by **30%**
- **Closed-loop Control:** Implemented **PID** w/ anti-windup for wire feed speed, **reducing bead variation by 30%**
- **Web UI:** Developed **TypeScript/React** command center with **Foxglove WebSocket**, migrated from rosbridge reducing crashes by **50%**; collaborated with weld engineers to build UI features improving weld process control

Relativity Space

Long Beach, CA

Robotics Software Engineer Intern

June 2023 - September 2023

- Exposing **OPC-UA** interface on Fronius welders to real-time context for Wire Arc Additive Manufacturing
- **ROS2 Driver:** Wrote an OPC-UA driver in **ROS2 (C++)** to read and write over RT (shared memory) and non-RT contexts, exposing **40+ new parameters** and increasing controllable welding parameters by **5x**
- **System Design:** Refactored welder state machine and inverted object dependencies, **reducing 1000+ lines of code**
- **UI Dev:** Created web interface for weld engineers to interact with OPC-UA parameters in **TypeScript**

Garmin

Brea, CA

Software Engineer Intern

June 2022 - December 2022

- Automating release processes and mock radio UI layer on the Tuner Team (Garmin AutoOEM Radio)
- **Process Automation:** Utilized Python REST APIs to automate page generation with user-configurable metadata
- **UI Automation:** Automated mock UI creation and deployment using MQTT & Node-RED for **214 D-Bus services**

InflammaSense

La Jolla, CA

Software Developer

January 2021 - September 2022

- Built medical device to record and analyze neural signals for early signs of sepsis, published **Scientific Reports** ([link](#))
- **Data Pipeline:** Designed a robust async data pipeline with the **ZeroMQ** Python API for the wireless display, transmission, storage (**PostgreSQL**), analysis of neural data for an array of **100+ medical devices (~155 Mbps)**
- **Project Management:** Documented a comprehensive system model and **leading a team of 5+ SWE** with Asana
- **UI Dev:** Built a GUI to display high throughput EKG data (8 kHz) w/ **PyQt**, enabling real-time data monitoring

PROJECTS

Autonomous Rover Path Planning

Yonder Dynamics

Software Lead

September 2020 - June 2023

- Lead software dev for rover to autonomously path and perform tasks in the University Rover Competition
- **Project Management:** Lead a team of **20+ SWE w/ SCRUM** and spearheaded documentation on Notion
- **Autonomous Exploration:** Implemented RRT* search in C++, **increasing rover exploration speed ~10x**
- **Obstacle Avoidance:** Created a **ROS** obstacle detection and avoidance pipeline in C++ with **RealSense** stereo camera to efficiently convert 3D pointclouds at 30Hz into occupancy grid and output avoidance waypoints

SKILLS

Programming Languages: Python, C++, Java, SQL, JS, PHP, HTML/CSS, TypeScript

Technologies: AWS, Git, Jupyter Notebook, Linux, Docker, Confluence, Jira, Gerrit, Gitlab CI

Frameworks: ROS2, OPC-UA, PyTorch, MQTT, Zenoh, Xenomai, NumPy, Pandas, SciKit, Ansible